

THE RULER-AND-COMPASS PLANE IS NOT 5-COLOURABLE

SERGI GALÁN

ABSTRACT. OpenAI recently proved that every colouring of the Euclidean plane with five colours gives the same colour to two points at distance 1, and formalized the proof in Lean. The proof works with the plane of algebraic points and assumes that every algebraic number of modulus 1 has roots of every order. We observe that square roots are enough. Hence the plane of points constructible with ruler and compass is not 5-colourable. These are the points of the smallest subfield of $\mathbb{C}$ closed under square roots. By the de Bruijn–Erdős theorem, some finite unit-distance graph with constructible vertices has no proper 5-colouring. Two observations suffice: a group-theoretic lemma about elements of order 2, and a choice of exponents in one multiple ergodic average that needs no shift. We checked the modified proof in Lean 4. The kernel accepts it, and only the standard axioms are used.

1. THE RESULT

The Hadwiger–Nelson problem asks for the least number $\chi(\mathbb{R}^2)$ of colours needed to colour the plane so that points at distance 1 get different colours. Isbell’s hexagonal colouring gives $\chi(\mathbb{R}^2) \leq 7$, and de Grey [4] proved $\chi(\mathbb{R}^2) \geq 5$; see Soifer [11] for the history. OpenAI [8] proved $\chi(\mathbb{R}^2) \geq 6$, with no regularity assumption on the colour classes. Their proof is not constructive, and we know of no finite unit-distance graph without a proper 5-colouring.

Identify the plane with $\mathbb{C}$. For a subset $S \subset \mathbb{C}$, a proper colouring of S is a colouring in which points of S at distance 1 get different colours. Let E_c be the field of *constructible numbers*: the smallest subfield of $\mathbb{C}$ closed under square roots, whose elements are the points constructible with ruler and compass from 0 and 1 [2, Ch. 10].

Theorem 1. *E_c has no proper 5-colouring.*

So the unit-distance graph on E_c has chromatic number 6 or 7. Larger planes contain E_c , so the theorem also holds for them. Examples are the origami numbers (closed under square and cube roots [1]), the numbers expressible by radicals, the algebraic numbers and $\mathbb{C}$ itself.

Corollary 2. *Some finite set of constructible points has no proper 5-colouring.*

Proof. A graph is 5-colourable if all its finite subgraphs are, by the de Bruijn–Erdős theorem [3]. $\square$

The coordinates of such a set lie in a field obtained from $\mathbb{Q}$ by finitely many square roots. We know of no such set. By contrast, the planes over $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ and $\mathbb{Q}(\sqrt{3}, \sqrt{11})$ have chromatic number 4 [9, 5].

OpenAI’s proof has two steps:

- [8, Prop. 4.2], the forward half of their transfer theorem, turns a proper colouring of the plane into a weak measurable colouring. Its proof first restricts the colouring to the countable plane E of algebraic points, and uses nothing else about it.
- [8, Thm. 1.4] shows that the plane has no weak measurable 5-colouring. It does not involve E .

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These results were obtained with the assistance of an AI system (Claude, by Anthropic), which modified the formal proof, ran the checks and helped to draft this note. The proof is OpenAI’s [8]; this note shows that it works over a smaller field. No mathematician has checked the note yet. The author is not a professional mathematician and submits it for independent verification.

We run the first step with E_c in place of the algebraic numbers. Section 2 lists what that step uses of its field. Sections 3 and 4 show that E_c has what it needs. Section 5 describes the formal proof.

Proof of Theorem 1. Suppose E_c had a proper 5-colouring. By Sections 2–4, the proof of [8, Prop. 4.2] goes through with E_c in place of the algebraic numbers. It gives a weak measurable 5-colouring of the plane, and [8, Thm. 1.4] excludes one. $\square$

2. WHAT THE PROOF USES OF THE FIELD

OpenAI work with $E = \overline{\mathbb{Q}} \subset \mathbb{C}$, its real points $F = E \cap \mathbb{R}$, and the group $K = \{u \in E : |u| = 1\}$ [8, Def. 2.1]. Their proof uses three properties of E :

- (E1) E is a countable subfield of $\mathbb{C}$, algebraic over $\mathbb{Q}$, closed under complex conjugation, and contains i . Since $\mathbb{Q}(i) \subset E$, E is dense in $\mathbb{C}$, F in $\mathbb{R}$ and K in the circle. These density facts are used in [8, Lemma 2.5, Lemma 4.1, Prop. 4.2].
- (E2) F is closed under square roots of its nonnegative elements. This is used in four places:
 - the decomposition $x = k + \bar{k}$ with $k = x/2 + i\sqrt{1 - x^2/4}$ in [8, Lemma 2.7];
 - the number $L = \sqrt{AB}$ in [8, Lemma 3.3];
 - the points $t + i\sqrt{1 - t^2}$ in [8, §3.5];
 - the moduli $|z| \in F$ of the elements $z \in E$.
- (E3) K is divisible. It is used in three places:
 - [8, Lemma 2.4] uses that a finite image of K is trivial;
 - [8, Lemma 3.2] uses that the power maps $u \mapsto u^n$ in its multiple averages are onto;
 - [8, Lemma 3.3] uses directly the square map, in (3.15), and $u \mapsto u^{-2}$, in (3.13).

For E_c , (E1) and (E2) hold, but (E3) holds only for the exponents $\pm 2^k$.

The map $u \mapsto u^2$ is onto K : the square roots of $a + bi$, with $a^2 + b^2 = 1$, have coordinates $\pm\sqrt{(1+a)/2}$ and $\pm\sqrt{(1-a)/2}$. The map $u \mapsto u^3$ is not onto. Take $u = (3+4i)/5 = (2+i)/(2-i)$. It has valuation 1 at the prime $2+i$ of $\mathbb{Z}[i]$, so it is not a cube in $\mathbb{Q}(i)$. Hence $z^3 - u$ is irreducible over $\mathbb{Q}(i)$, and its roots, of degree 3 over $\mathbb{Q}(i)$, are not constructible.

3. ELEMENTS OF ORDER TWO

In [8, Lemma 2.4], a coset $d + C$ of positive mass is fixed by some rotation $u \neq 1$. Then $(u-1)d \in C$, and so $d \in C$. The formalization takes $u = -1$. That -1 fixes the coset follows from the next lemma, applied to the action of K on the finite orbit of the coset.

Lemma 3. *Let G be a commutative group in which every element is a square, and let $f: G \rightarrow H$ be a homomorphism with finite image. Then $f(g) = 1$ for every $g \in G$ with $g^2 = 1$.*

Proof. Squaring maps the finite group $f(G)$ onto itself, because every element of G is a square. So squaring is injective on $f(G)$. Since $f(g)^2 = f(g^2) = 1 = 1^2$, it follows that $f(g) = 1$. $\square$

The same lemma also gives the second use of divisibility in the formal proof. In the null-set statement for lines in [8, Lemma 2.4], it shows that K is infinite: if K were finite, the identity map of K would send -1 to 1.

4. THE TRIPLE AVERAGE AT THE EXPONENTS $-1, 0, 1$

We use the notation of [8, §3]:

- X is an ergodic standard probability-preserving K -system, and Y is a factor with $R = X \times_Y X$ ergodic, as in [8, Lemma 3.2];
- $P = \mathbb{E}(\cdot | Y)$;
- U_u is the Koopman operator of $u \in K$. It is unitary, multiplicative, and commutes with P ;
- averages are taken along a Følner sequence of K .

The radial inequality $m(A)m(B) \leq m(|A - B|)$ of [8, Lemma 3.3] rests on two facts:

- the identity $|A - Bu^2| = |-L + (A - B)u + Lu^2|$, for $L = \sqrt{AB}$ and $|u| = 1$;

- the multiple average of [8, Lemma 3.2] for the characters $\psi_{-L} \cdot U_u \psi_{A-B} \cdot U_{u^2} \psi_L$, at the exponents 0, 1, 2.

In the proof of [8, Lemma 3.2], at (3.9), a common integer is added to the exponents of each term so that none is zero. The formalization uses one shift for all terms and requires every shifted exponent to be onto. For 0, 1, 2, one of three consecutive nonzero integers is a nonzero multiple of 3, so this needs cube roots. A shift chosen separately for each term would avoid them; we avoid the shift altogether.

Since $|u| = 1$,

$$|-L + (A - B)u + Lu^2| = |-Lu^{-1} + (A - B) + Lu|,$$

and, since U_u is unitary, multiplicative and commutes with P ,

$$\|P(a \cdot U_u b \cdot U_{u^2} c)\| = \|P(U_{u^{-1}} a \cdot b \cdot U_u c)\|.$$

So it is enough to prove the conclusion (3.2) of [8, Lemma 3.2] at the exponents $-1, 0, 1$, where the middle factor is fixed.

Lemma 4. *Let X, Y and R be as above, and suppose that $u \mapsto u^2$ is onto K . For bounded functions a, b, c on X ,*

$$\text{Avg}_u \|P(U_{u^{-1}} a \cdot b \cdot U_u c) - U_{u^{-1}} P a \cdot P b \cdot U_u P c\|_2^2 \longrightarrow 0$$

along every Følner sequence.

Proof. Write each factor as $v = Pv + (v - Pv)$ and expand. The term with no centred factor is $U_{u^{-1}} P a \cdot P b \cdot U_u P c$. It is Y -measurable, so P fixes it.

Consider a term whose only centred factor is $b - Pb$. Its other two factors are Y -measurable, so its projection is $U_{u^{-1}} P a \cdot U_u P c \cdot P(b - Pb) = 0$.

In every remaining term a centred factor sits at the exponent -1 or 1 . Let $W_u = U_{u^{-1}} a' \cdot b' \cdot U_u c'$ be such a term. By conditional independence,

$$\|PW_u\|_2^2 = \int_R \Phi \cdot G_u d\mu_R, \quad \Phi = b' \otimes \bar{b}', \quad G_u = U_{u^{-1}}(a' \otimes \bar{a}') \cdot U_u(c' \otimes \bar{c}').$$

One of the two factors of G_u is centred over Y , because $\mathbb{E}(w \otimes \bar{w} \mid Y) = |Pw|^2$.

Statement (3.5) of [8], applied on R to the two distinct nonzero exponents -1 and 1 , says that $\text{Avg}_u G_u \rightarrow 0$ in $L^2(R)$. Its proof uses three facts:

- R is ergodic;
- the pair criterion (3.3) holds on R at $d = \pm 1$;
- the case $l = 1$ holds through the power maps $u \mapsto u^{\pm 2}$, which are onto.

The function Φ does not depend on u , so $|\text{Avg}_u \int_R \Phi G_u d\mu_R| \leq \|\Phi\|_2 \|\text{Avg}_u G_u\|_2 \rightarrow 0$. This is where it matters that the middle factor is fixed. There are finitely many terms, and the lemma follows. $\square$

The power maps used are $u \mapsto u^{\pm 1}$ and $u \mapsto u^{\pm 2}$. The other two averages in [8, Lemma 3.3] need only the square map:

- the pair at the exponents 0, 2 is the pair criterion (3.3) with $d = 2$;
- the pair at the exponents 1, -1 needs no shift, and its difference is 2.

With Lemmas 3 and 4, the rest of the proof of [8, Prop. 4.2] goes through over E_c unchanged. This completes the proof of Theorem 1.

5. THE FORMAL PROOF

OpenAI's formalization [8] uses Lean 4 [6] with Mathlib [7]; we used the same versions, Lean 4.34.1 and Mathlib at commit `d13f23b7`. The proof is the import closure of their file `Five.lean`: 70 modules of their library, on top of Mathlib. We changed it in two ways:

- a new file defines the field. It proves that the field is algebraic and closed under conjugation, contains i , and satisfies (E2). Countability and the square roots in K are derived from these facts in other files. Eleven files of the closure were adapted.

- Two new lemmas in `Energy.lean` prove Lemma 4, and the radial inequality uses it. Like the rest of the formal proof, they are stated for invariant means rather than Følner sequences, so Lean proves the invariant-mean form of Lemma 4.

The final statement writes the field out, so the reader does not need to trust our definitions. In abbreviated form:

```
theorem sqrtClosure_plane_not_five_colourable :
  ¬ ∃ c : (sInf {s : IntermediateField ℚ ℂ | ∀ x : ℂ, x² ∈ s → x ∈ s}) → Fin 5,
    ∀ x y, ||(x : ℂ) - (y : ℂ)|| = 1 → c x ≠ c y
```

`#print axioms` lists only `propext`, `Classical.choice` and `Quot.sound`. A second file proves Corollary 2 from it, with Mathlib’s compactness theorem for graph homomorphisms; the axioms are the same.

We also checked the proof in two further ways:

- a fresh copy of OpenAI’s sources (commit `fd4aeeb2`) was patched and built from scratch: the 72 modules of the closure, namely their 70, the field file and the statement file, with none failing. Mathlib was prebuilt;
- `leanchecker` replayed these 72 modules in the kernel. Mathlib was not replayed.

An AI-assisted review of an earlier version of these changes found no mathematical error.

Before the constructible case we built the origami and radical cases and checked their axioms. The origami case was also rebuilt from scratch with the published script. The patches, the statements and a build script are in `lean/external/openai-five/fields/` of [10], with a copy of OpenAI’s licence (Apache 2.0).

6. REMARKS

Property (E2) says that F is a Euclidean field, and the method needs it: with $B = 1$, [8, Lemma 3.3] uses $\sqrt{A}$ for every $A \geq 0$ in F . Every Euclidean subfield of $\mathbb{R}$ contains the real constructible numbers. So E_c is the smallest field of the form $F(i)$ to which the proof applies in this form.

Two questions remain open:

- Does the plane over the Pythagorean closure of $\mathbb{Q}$ have a proper 5-colouring? There K is still 2-divisible, because $(1 \pm a)/2 = ((1 \pm a)/2)^2 + (b/2)^2$ is a sum of squares. So only (E2) fails.
- How many square roots must be adjoined to $\mathbb{Q}$ to obtain a field whose plane has no proper 5-colouring? Corollary 2 says that finitely many suffice.

An explicit finite unit-distance graph with no proper 5-colouring would bound the answer to the second question. It would also give an explicit witness for $\chi(\mathbb{R}^2) \geq 6$.

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